

# AI Models in Autonomous Game Development

How AI handles game development from scratch to release and what are the costs involved?

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# The Costs of Generative AI

**Pricing**

[Copy Page](#)

**Flagship models**

Our latest models

Prices per 1M tokens.

**Standard** Batch Flex Priority

| Model        | Short context |              |          | Long context |              |         |
|--------------|---------------|--------------|----------|--------------|--------------|---------|
|              | Input         | Cached input | Output   | Input        | Cached input | Output  |
| gpt-5.4      | \$2.50        | \$0.25       | \$15.00  | \$5.00       | \$0.50       | \$22.50 |
| gpt-5.4-mini | \$0.75        | \$0.075      | \$4.50   | -            | -            | -       |
| gpt-5.4-nano | \$0.20        | \$0.02       | \$1.25   | -            | -            | -       |
| gpt-5.4-pro  | \$30.00       | -            | \$180.00 | \$60.00      | -            | -       |

<https://claude.com/pricing#api>

<https://developers.openai.com/api/docs/pricing?latest-pricing=standard>

**Latest models**

[Contact sales](#) [Start building](#)

|                                                                 |                                                                       |                                                        |
|-----------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------|
| <b>Opus 4.6</b><br>Most intelligent model for agents and coding | <b>Sonnet 4.6</b><br>Optimal balance of intelligence, cost, and speed | <b>Haiku 4.5</b><br>Fastest, most cost-efficient model |
| Input                                                           | Input                                                                 | Input                                                  |
| Output                                                          | Output                                                                | Output                                                 |
| Prompt caching                                                  | Prompt caching                                                        | Prompt caching                                         |
| Write                                                           | Write                                                                 | Write                                                  |
| Read                                                            | Read                                                                  | Read                                                   |

# LLMs in SWE

Text Arena  Coding

View overall rankings across various AI models in text-to-text tasks across math, coding, creative writing, and other open-ended domains.

🕒 Mar 31, 2026 🗳️ 1,015,744 votes 📄 332 models

⚙️ Hide Filters 🔍 📄

| Rank ↕ | Rank Spread ↕ | Model ↕                                                                                                                                                      | Score ↓  | Votes ↕ | Price \$/M ↕  | Context ↕ |
|--------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|---------------|-----------|
| 1      | 1 → 3         |  <b>claude-opus-4-6-thinking</b><br>Anthropic · Proprietary                 | 1556 ±11 | 3,017   | \$5 / \$25    | 1M        |
| 2      | 1 → 7         |  <b>claude-opus-4-6</b><br>Anthropic · Proprietary                          | 1550 ±10 | 3,503   | \$5 / \$25    | 1M        |
| 3      | 2 → 16        |  <b>gemini-3.1-pro-preview</b><br>Google · Proprietary                      | 1534 ±10 | 4,057   | \$2 / \$12    | 1M        |
| 4      | 1 → 23        |  <b>gpt-5.4-high</b><br>OpenAI · Proprietary                                | 1533 ±15 | 1,596   | \$2.50 / \$15 | 1.1M      |
| 5      | 3 → 17        |  <b>claude-opus-4-5-20251101-thinking-32k</b><br>Anthropic · Proprietary    | 1531 ±7  | 7,677   | \$5 / \$25    | 200K      |
| 6      | 2 → 28        |  <b>grok-4.20-beta1</b><br>xAI · Proprietary                                | 1526 ±15 | 1,479   | N/A           | N/A       |
| 7      | 2 → 27        |  <b>grok-4.20-beta-0309-reasoning</b><br>xAI · Proprietary                  | 1525 ±14 | 1,698   | \$2 / \$6     | 2M        |
| 8      | 3 → 29        |  <b>claude-sonnet-4-6</b><br>Anthropic · Proprietary                        | 1522 ±12 | 2,554   | \$3 / \$15    | 1M        |
| 9      | 3 → 29        |  <b>gpt-5.2-chat-latest-20260210</b><br>OpenAI · Proprietary                | 1520 ±11 | 3,010   | \$1.75 / \$14 | 128K      |
| 10     | 3 → 25        |  <b>claude-sonnet-4-5-20250929-thinking-32k</b><br>Anthropic · Proprietary | 1519 ±6  | 12,392  | \$3 / \$15    | 200K      |

<https://arena.ai/leaderboard/text/coding>

# The Carbon Footprint of a Prompt

- Training vs. Inference – where does the power go?
- Global grid average: ~400g CO<sub>2</sub> per consumed kWh.
- Data Center PUE (Power Usage Effectiveness) – industry standard is often 1.20.
- The impact of model complexity on energy consumption.

“Microsoft will restart a nuclear power plant in the US. Power is needed for artificial intelligence.”

**Dla Microsoft w USA uruchomią ponownie elektrownię atomową. Trzeba prądu dla sztucznej inteligencji**

ENERGETYKA JĄDROWA 23.09.2024, 05:50

Andrzej Kublik

**wyborcza.biz**



• Elektrownia atomową Three Mile Island w Pensylwanii w USA (Fot. Shutterstock)

# The Thirst of AI - Water Consumption

- Why do Data Centers "drink" water?
- Evaporative cooling towers.
- Estimated consumption: ~2.0 Liters of water per kWh.
- Local impact - how server farms affect regional water resources.



The screenshot shows the top portion of a news article from Chinanews. The header is dark blue with the 'CHINADAILY' logo in white, followed by 'HONG KONG' and a dropdown arrow. To the right is the logo of The Hong Kong University of Science and Technology. Below the header is a navigation bar with links: HOME, NEWS (with a dropdown arrow), FOCUS (underlined), OPINION (with a dropdown arrow), GBA (with a dropdown arrow), and VIDEO (with a dropdown arrow). Below the navigation bar, the breadcrumb trail reads 'FOCUS > In-depth China > Content'. The article's publication information is 'Published: 04:36, January 12, 2026', followed by a 'PDF View' button. The main headline is 'Lake water tames data center heat' in a large, bold, black font.

**CHINADAILY** | HONG KONG ▼

THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

HOME NEWS ▼ FOCUS ▼ OPINION ▼ GBA ▼ VIDEO ▼

FOCUS > In-depth China > Content

Published: 04:36, January 12, 2026 [PDF View](#)

## Lake water tames data center heat

# Vibe Coding vs. Sustainable Coding

- The "Vibe Coding" phenomenon – rapid prototyping without optimization.
- A bad prompt = wasted money and resources.
- "Let's throw AI at it" – is it always worth using LLMs for simple problems?
- The need for optimization and precise Game Design Documents (GDD).

"An engineer at OpenAI processed 210 billion tokens last week – enough text to fill Wikipedia 33 times.

At Anthropic, one user ran up a \$150,000 Claude Code bill in a single month."

<https://www.nytimes.com/2026/03/20/technology/token-maxxing-ai-agents.html>



# Is Local/Small AI the Future?

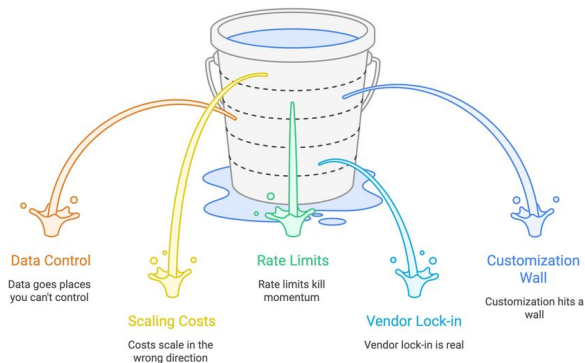
- Small models running locally (e.g., Raptor mini).
- Zero API cost and no network latency.
- Leveraging the creator's machine power instead of the cloud.
- The trade-off between model size and coding capabilities

## Realistic hardware tiers

| Setup      | Hardware                 | Models                          | Cost                       |
|------------|--------------------------|---------------------------------|----------------------------|
| Starter    | RTX 3060 12GB, 32GB RAM  | 7B models (Mistral, Llama 3 8B) | ~\$1,200-1,500             |
| Mid-tier   | RTX 4090 24GB, 64GB RAM  | 7B-30B models, quantized 70B    | ~\$2,500-3,500             |
| Production | Multi-GPU or cloud A100s | 70B+ models, high concurrency   | \$10,000+ or cloud pricing |

<https://blog.prem.ai/self-hosted-ai-models-a-practical-guide-to-running-llms-locally-2026/>

## API-Based AI Challenges



# Self-Hosted AI vs API Services: Cost Breakdown

## API costs at scale

| Monthly usage | GPT-4o (~\$5/1M tokens) | GPT-4o-mini (~\$0.60/1M tokens) | Claude Sonnet (~\$3/1M tokens) |
|---------------|-------------------------|---------------------------------|--------------------------------|
| 100K tokens   | \$0.50                  | \$0.06                          | \$0.30                         |
| 1M tokens     | \$5                     | \$0.60                          | \$3                            |
| 10M tokens    | \$50                    | \$6                             | \$30                           |
| 100M tokens   | \$500                   | \$60                            | \$300                          |
| 1B tokens     | \$5,000                 | \$600                           | \$3,000                        |

## Self-hosted costs

| Setup                        | Hardware cost | Monthly running cost | Models supported           |
|------------------------------|---------------|----------------------|----------------------------|
| Starter (RTX 3060 12GB)      | ~\$1,200      | ~\$30 electricity    | 7B models                  |
| Mid-tier (RTX 4090 24GB)     | ~\$2,500      | ~\$50 electricity    | 7B-30B, quantized 70B      |
| Production (2x RTX 4090)     | ~\$5,500      | ~\$100 electricity   | 70B models comfortably     |
| Enterprise (A100 80GB cloud) | \$0           | ~\$2,500/month       | Any size, high concurrency |

## Breakeven analysis

| Your monthly API spend | Breakeven timeline (mid-tier setup) |
|------------------------|-------------------------------------|
| \$100/month            | ~25 months                          |
| \$300/month            | ~8 months                           |
| \$500/month            | ~5 months                           |
| \$1,000/month          | ~2.5 months                         |
| \$2,000+/month         | Under 2 months                      |



# Open-Source Models Worth Self-Hosting

**Llama 3 (Meta)**

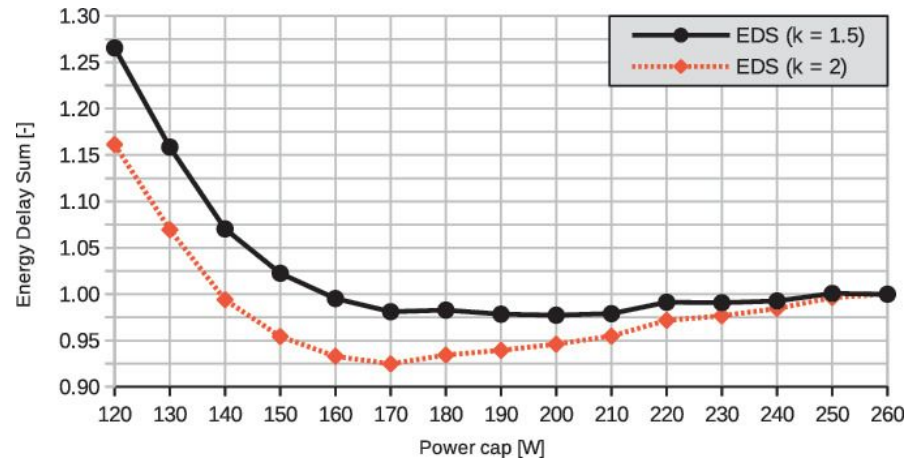
**Mistral (Mistral AI)**

**DeepSeek R1**

**Qwen 2.5 (Alibaba)**

# Dynamic GPU power capping

The DEPO tool achieved energy savings of up to 22–23% on NVIDIA A100 and RTX A4500 GPUs across CNN training workloads (ResNet152, DenseNet161, VGG-19) and GEMM benchmarks, with bi-objective metrics (EDP/EDS) enabling **~15–18% energy savings** at less than **10% performance loss** in most cases.



<https://www.sciencedirect.com/science/article/pii/S0167739X23001267>  
#fig1

# Estimating the Unseen

- Exact architecture, hardware usage, and energy consumption of top-tier models remain strictly confidential.
- The environmental and infrastructural results in our experiment are estimations, not direct measurements.
- Methodology: Calculations are based on data from open-source models and fragmented industry reports.
- Despite being approximations, they provide a crucial, realistic perspective on the scale of AI's impact.

# Our Experiment: Chess Developement

For the purpose of this project, we conducted an experiment using five AI models available in Github Copilot student plan:

1. Claude Haiku 4.5,
2. Grok Code Fast 1
3. GPT-5.3-Codex
4. Gemini 3.1 Pro
5. Raptor mini

Their task was to code a Chess game entirely without our manual intervention.

# Methodology Outline

As part of our methodology, we prepared an environment with PyGame library installed, proper assets available, and a **Game Design Document (GDD)** outlining the game's concept, rules, core mechanics, and artistic vision, along with a few technical details like asset paths and element dimensions.

Next, each model was given the exact same initial prompt, followed by simple, structured prompts detailing any bugs that needed fixing. All models were accessed and utilized through the GitHub Copilot interface within VS Code.

All of this was done to ensure maximum repeatability and objectivity throughout the process.

# Prompts Structure

## Initial:

Using the Game Design Document [GDD.md](#) and your own knowledge, create a Chess game in single file [Chess-{model\\_name}.py](#) with use of PyGame library available in .venv. Make sure to include every mechanic and visual design that is described in GDD.

1. For initial state check especially: [Setup](#), [Visuals](#) and [UI panel](#).
2. For mechanics and game flow check especially: [Rules](#) and [Course of the game](#).

## Following:

Fix the issues:

1. Issue nr 1
2. Issue nr 2
3. ...



# Results

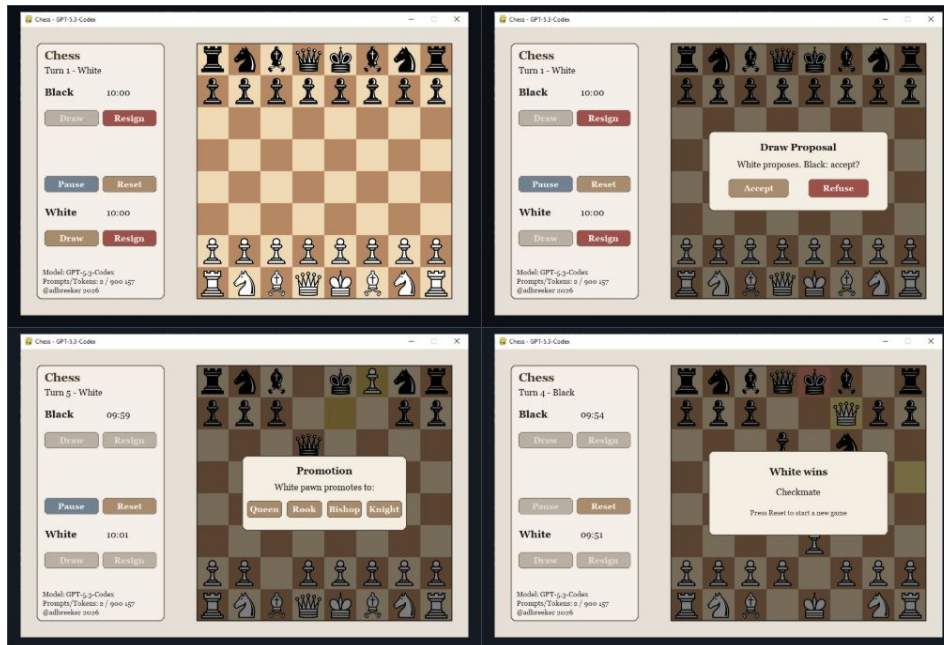
The experiment resulted in five fully functional chess games, featuring a working interface and both basic and advanced chess mechanics. Each game also includes a simple engine capable of detecting legal moves, checks, and checkmates.

While four out of the five models successfully completed the task in under 20 prompts, the remaining 5th model's work was halted after the 20th prompt. To finalize the project, this model's code received its final polishes from the top-performing model in our lineup.

Throughout each experiment, we also collected specialized performance metrics for further evaluation, such as total token usage and the number of tool calls made by each model.

# GPT-5.3-Codex

- Prompts: 2
- Total Tokens: 900,157
- Model Turns: 16
- Tool Calls: 19
- Total Events: 55





# Gemini 3.1 Pro

- Prompts: 3
- Total Tokens: 1,123,267
- Model Turns: 25
- Tool Calls: 25
- Total Events: 68



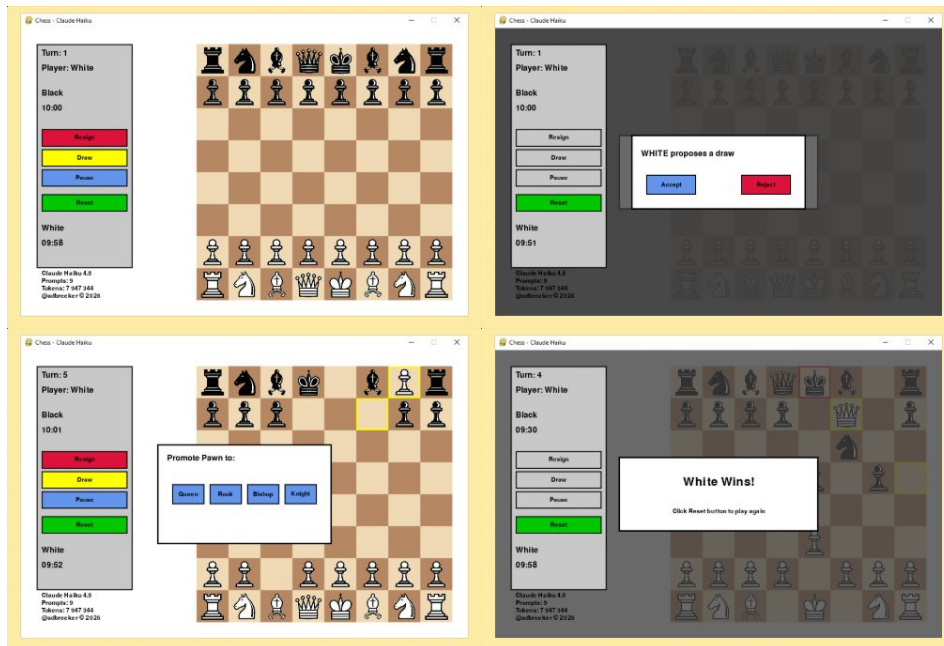
# Grok Code Fast 1

- Prompts: 6
- Total Tokens: 3,430,780
- Model Turns: 64
- Tool Calls: 57
- Total Events: 157



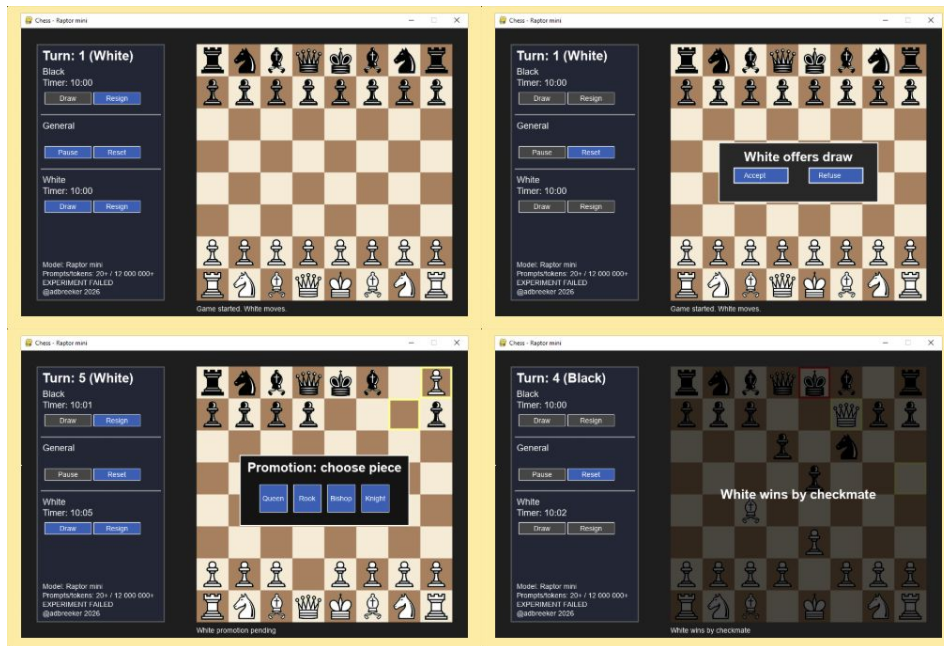
# Claude Haiku 4.5

- Prompts: 9
- Total Tokens: 7,947,344
- Model Turns: 96
- Tool Calls: 107
- Total Events: 280



# Raptor mini

- Prompts: 20+
- Total Tokens: 12,000,000+
- Model Turns: 140
- Tool Calls: 139
- Total Events: 415



# Costs

| Model            | Total Tokens | Est. Total Energy (Wh)* | Est. Carbon (gCO <sub>2</sub> e)** | Est. Water (ml)*** | Est. API Cost (USD) |
|------------------|--------------|-------------------------|------------------------------------|--------------------|---------------------|
| GPT-5.3-Codex    | 900,157      | 90.0 Wh                 | 36.0 g                             | 180 ml             | \$9.00              |
| Gemini 3.1 Pro   | 1,123,267    | 112.3 Wh                | 44.9 g                             | 224 ml             | \$7.86              |
| Grok Code Fast 1 | 3,430,780    | 343.0 Wh                | 137.2 g                            | 686 ml             | \$13.72             |
| Claude Haiku 4.5 | 7,947,344    | 794.7 Wh                | 317.8 g                            | 1,589 ml           | \$3.97              |
| Raptor mini      | >12,000,000  | >1,200.0 Wh             | >480.0 g                           | >2,400 ml          | N/A (Local/Free)    |

\*Includes estimated 1.20 Data Center PUE. \*\*Calculated using the global grid average of 400g CO<sub>2</sub>/kWh [10]. \*\*\*Estimated at 2.0 Liters per kWh for cooling tower evaporation.

# Conclusions

## Top 2

Gemini 3.1 Pro:

- Recovered well from initial struggles with low amount of prompts (3)
- Final result is good with proper visuals
- Mechanical soundness achieved quickly
- Tool usage improved after guidance
- Cost-effective with minimal prompts needed

## Top 1

GPT-5.3-Codex:

- Delivered the best overall result with only 2 prompts
- Outstanding visual and mechanical implementation
- Creative features not conceived by other models
- Most efficient token-to-quality ratio
- Despite long response times, the outcome justifies the investment

## Top 3

Claude Haiku 4.5:

- Consistently responsive and reliable
- 2-3x cheaper than top models while delivering comparable quality
- Good balance of speed, cost, and mechanical/visual quality
- Handles corrections well
- Good for iterative development

# THE END

## Authors:

- Adam Cherek, s207250 – experiment and analysis of the results
- Mateusz Nowak, s189420 – research and analysis of the results
- Paweł Mańczak, s188756 – research and analysis of the results

Thank you for your attention and for your time. We hope you found our presentation interesting and engaging. If you'd like to explore our work personally, all the games we created, along with the experiment results and project documentation, are available via the QR code.

